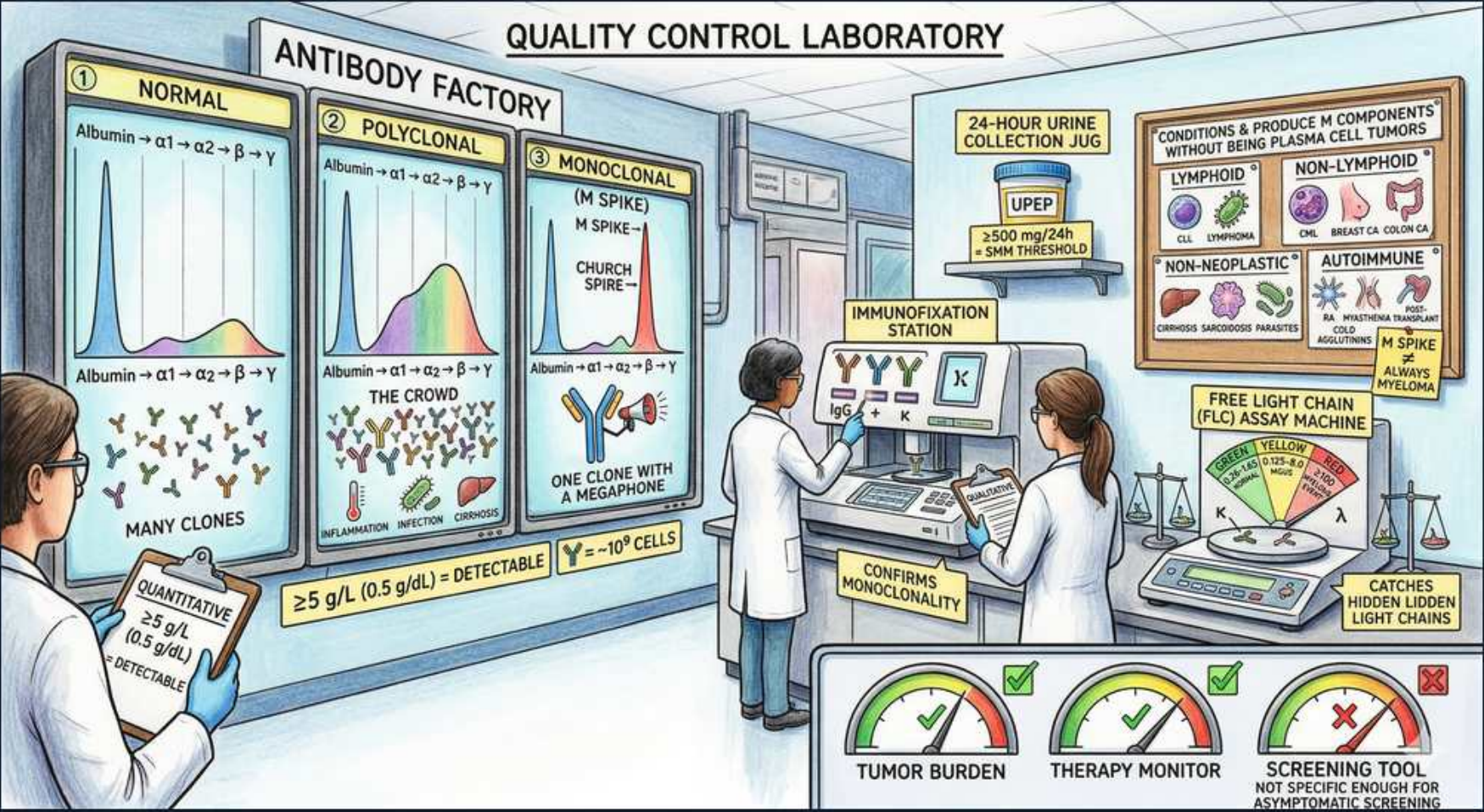


Paraprotein Lab Workup — SPEP, Immunofixation & Free Light Chains



1. Normal serum protein electrophoresis

WHAT YOU SEE

Panel 1: smooth albumin→α→β→γ curve.

WHAT IT ENCODES

Normal SPEP shows a tall albumin peak then a broad, low polyclonal gamma region. Serves as the baseline against which spikes are read.

BOARD TRAP

A normal gamma region is broad and low — any tall, narrow gamma peak is the abnormality to chase.

2. Polyclonal gammopathy

WHAT YOU SEE

Panel 2: broad 'hill' in gamma zone, many clones.

WHAT IT ENCODES

A broad-based gamma elevation = polyclonal, from many B-cell clones. Seen in infection, inflammation, chronic liver disease, autoimmune disease — NOT a malignant plasma-cell clone.

BOARD TRAP

A broad polyclonal hump is reactive/benign; only a sharp narrow spike (monoclonal) suggests plasma-cell malignancy.

3. Monoclonal spike (M-protein)

WHAT YOU SEE

Panel 3: sharp 'church spire' spike, one clone.

WHAT IT ENCODES

A monoclonal spike is a tall, narrow band ('church spire') from a single plasma-cell clone making identical immunoglobulin. Detectable on SPEP when ≥0.5 g/dL and ~10^9 cells.

BOARD TRAP

One narrow spike = monoclonal = clonal plasma-cell process; the spire shape, not the height alone, defines monoclonality.

4. Immunofixation — confirms monoclonality

WHAT YOU SEE

Immunofixation station with antisera lanes.

WHAT IT ENCODES

Immunofixation identifies the exact heavy- and light-chain class (e.g., IgG-kappa) and confirms a band is truly monoclonal. More specific than SPEP, which only quantifies.

BOARD TRAP

SPEP detects/quantifies the spike; immunofixation TYPES it. You need IFE to confirm clonality and class — SPEP alone isn't enough.

5. 24-hour urine / UPEP (Bence Jones)

WHAT YOU SEE

24-hour urine collection jug, UPEP.

WHAT IT ENCODES

Urine studies detect free light chains (Bence Jones protein). A 24-hour UPEP quantifies urinary M-protein; a threshold of ~500 mg/24h is a meaningful marker for light-chain disease.

BOARD TRAP

Standard urine dipstick misses Bence Jones light chains — you need UPEP/immunofixation on a 24-hour collection.

6. Free light chain (FLC) assay

WHAT YOU SEE

FLC assay machine, kappa/lambda dial.

WHAT IT ENCODES

The serum free light-chain assay measures kappa and lambda; an abnormal kappa:lambda RATIO signals clonality. Catches 'light-chain only' and oligosecretory/nonsecretory disease missed by SPEP.

BOARD TRAP

FLC ratio (not absolute level) defines clonality, and it catches light-chain-only myeloma that SPEP/IFE can miss.

7. Conditions making M-protein without PC tumor

WHAT YOU SEE

List of non-plasma-cell M-protein causes.

WHAT IT ENCODES

Lymphoid (CLL, lymphoma), non-neoplastic (chronic infection, autoimmune), and other clonal disorders can produce M-components without being a plasma-cell tumor. Context dictates work-up.

BOARD TRAP

An M-protein does NOT equal myeloma — lymphoproliferative and reactive conditions also make monoclonal proteins.

8. M-spike threshold for detection

WHAT YOU SEE

Note: ≥0.5 g/dL detectable, ~10^9 cells.

WHAT IT ENCODES

SPEP reliably detects M-proteins at ≥0.5 g/dL, corresponding to roughly 10^9 clonal cells. Smaller clones need the more sensitive FLC assay or immunofixation.

BOARD TRAP

A 'negative' SPEP doesn't exclude clonal disease below ~0.5 g/dL — escalate to IFE/FLC when suspicion is high.

9. Tumor burden marker

WHAT YOU SEE

Gauge: tumor burden.

WHAT IT ENCODES

The M-protein quantity tracks tumor burden and is used serially to monitor disease and therapy response in myeloma and related disorders.

BOARD TRAP

Use the SAME measured M-protein (serum spike or involved FLC) consistently for monitoring — don't switch metrics mid-course.

10. Therapy monitoring

WHAT YOU SEE

Gauge: therapy monitor.

WHAT IT ENCODES

Falling M-protein/FLC indicates response; rising values indicate progression. Response criteria depend on serial quantitative measurement.

BOARD TRAP

Response is judged on serial quantitative change, not a single value; nonsecretory disease must be followed by FLC or imaging.

11. Screening limitation

WHAT YOU SEE

Gauge: screening tool — not specific.

WHAT IT ENCODES

These assays are diagnostic/monitoring tools, not population screening tests — they're not specific enough to screen asymptomatic people for plasma-cell disease.

BOARD TRAP

Don't use SPEP/FLC to screen asymptomatic patients; order them when there is a clinical indication (CRAB, neuropathy, etc.).

12. IgM threshold context

WHAT YOU SEE

Note: ≥ 500 mg/24h, ≥ 3 g/dL IgM thresholds.

WHAT IT ENCODES

Specific thresholds (e.g., urinary light chain ≥ 500 mg/24h, or IgM spike levels) help stratify monoclonal gammopathies and trigger further work-up.

BOARD TRAP

Thresholds differ by isotype and fluid — apply the correct cutoff for serum vs 24-hour urine, IgG vs IgM.

13. Kappa vs lambda restriction

WHAT YOU SEE

Antibody scales weighing kappa vs lambda.

WHAT IT ENCODES

A clone overproduces ONE light chain, skewing the kappa:lambda ratio. The skew direction identifies the involved clone and is central to FLC interpretation.

BOARD TRAP

It's the RATIO skew (not just a high absolute light chain, which also rises in renal failure) that indicates clonality.

14. Quantitative vs qualitative methods

WHAT YOU SEE

Technician with quantitative clipboard.

WHAT IT ENCODES

SPEP gives quantitative spike size; immunofixation and FLC give qualitative/typing and sensitive detection. Together they define presence, type, and burden.

BOARD TRAP

Quantitation (SPEP) and characterization (IFE) answer different questions — boards expect you to order both appropriately.

15. One clone with a megaphone

WHAT YOU SEE

Single antibody shouting through a megaphone.

WHAT IT ENCODES

The visual mnemonic: a single clone loudly overproducing identical immunoglobulin = monoclonal gammopathy. Contrasts with the quiet, diverse polyclonal crowd.

BOARD TRAP

Monoclonality = one loud identical product; polyclonal = many different quiet ones. The shape on SPEP reflects this.